

SKILLS GUIDE



Excellent communication skills for collaborating with colleagues on a range of projects.



The creativity and innovation necessary to find working solutions to engineering problems.



The ability to handle pressure while maintaining good working relationships.



Good computer skills to work with Computer-aided Design (CAD) programs.



A strong eye for detail and the ability to build and test working prototypes.



MATERIALS ENGINEER

Develops and tests the properties of materials, such as their strength or resistance to corrosion, to see if they're fit for a specific purpose.



INDUSTRIAL PRODUCTION

MANAGER Refines mechanical systems and deals with on-site problems that may arise at manufacturing facilities and production lines.



MINING ENGINEER Manages the safe operation of mechanized wells and mines for the efficient extraction of oil and minerals.

RELATED CAREERS

- ▶ **AEROSPACE ENGINEER** *see pp. 190–191*
- ▶ **AUTOMOTIVE ENGINEER** Works in production plants, designing and manufacturing road vehicles. An automotive engineer may also build race cars or other specialized vehicles.
- ▶ **BUILDING SERVICES ENGINEER** Designs, installs, and maintains machinery—such as heating, lighting, and plumbing systems—in buildings. Apprenticeships may be available to those with a high-school diploma.
- ▶ **MECHATRONIC ENGINEER** Develops products by combining mechanical, electronic, and computer components. These products include home appliances, cameras, and computer hard drives.

AT A GLANCE



YOUR INTERESTS Engineering • Science • Mathematics • Physical sciences • Design • Computers



ENTRY QUALIFICATIONS

A degree in mechanical engineering is a minimum requirement; some employers require higher degrees.



LIFESTYLE Regular hours are the norm, though engineers in some sectors may need to travel or work overnight to meet project deadlines.



LOCATION Although the work is mainly office based, engineers may need to make frequent visits to manufacturing and testing sites.



THE REALITIES Higher education in mechanical engineering is notoriously tough. Ongoing learning is essential to keep up to date with new technologies.